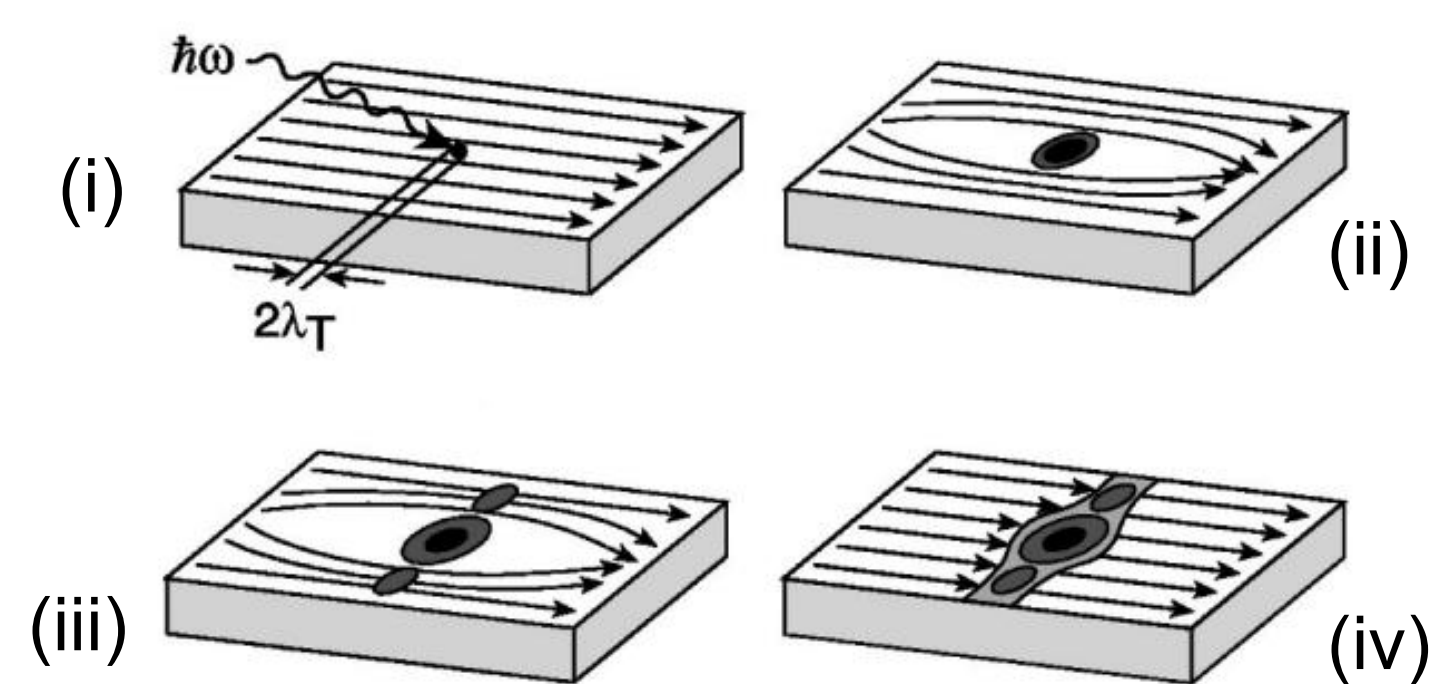


Modelling of Multi-Photon Events of a Superconducting Single-Photon Nanostripe Detector for Photon Number Resolution Readout Circuit

Introduction

Superconducting nanostripe single photon detectors (SNSPDs) [1] exhibit near-unit quantum efficiency [2,3], gigahertz counting rates [4], picosecond jitter [5], sub-hertz dark counts [6], and a relatively high operating temperature (2-4 K). Their operation is schematically shown in FIG. 1.



- (i) A superconductor nanostripe is maintained at a temperature far below its critical temperature T_c and biased with a dc current I_b just below the critical current I_c .
- (ii) After an optical photon is absorbed, a hotspot is created.
- (iii) The hotspot expands, forcing I_b to flow along its periphery.
- (iv) As the hotspot continues to grow, a resistive barrier is formed across the width of the nanostripe.

FIG. 1: Schematics of SNSPD photon detection.

The SNSPD operates typically in a threshold mode, and photon number resolution (PNR) is a challenging task. We demonstrate that in a single meander, high kinetic inductance L_k SNSPD, a limited PNR can be achieved by tracing the rising-edge-time of the SNSPD photopresponse. A single-flux-quantum (SFQ) readout scheme is incorporated to yield separated SFQ timing pulses for different photon events.

Design Objectives:

1. Model SNSPD multi-photon dynamics. Simulate 0, 1, and ≥ 2 absorbed-photon events to capture photoresponse risetime differences.
2. Implement SFQ timing-based readout. Convert rise-time variations into SFQ bit patterns: '00' = no photon, '01' = 1 photon, '10' = ≥ 2 photons, for logic states (0), (1), and (2), respectively.

SNSPD Dynamic Model

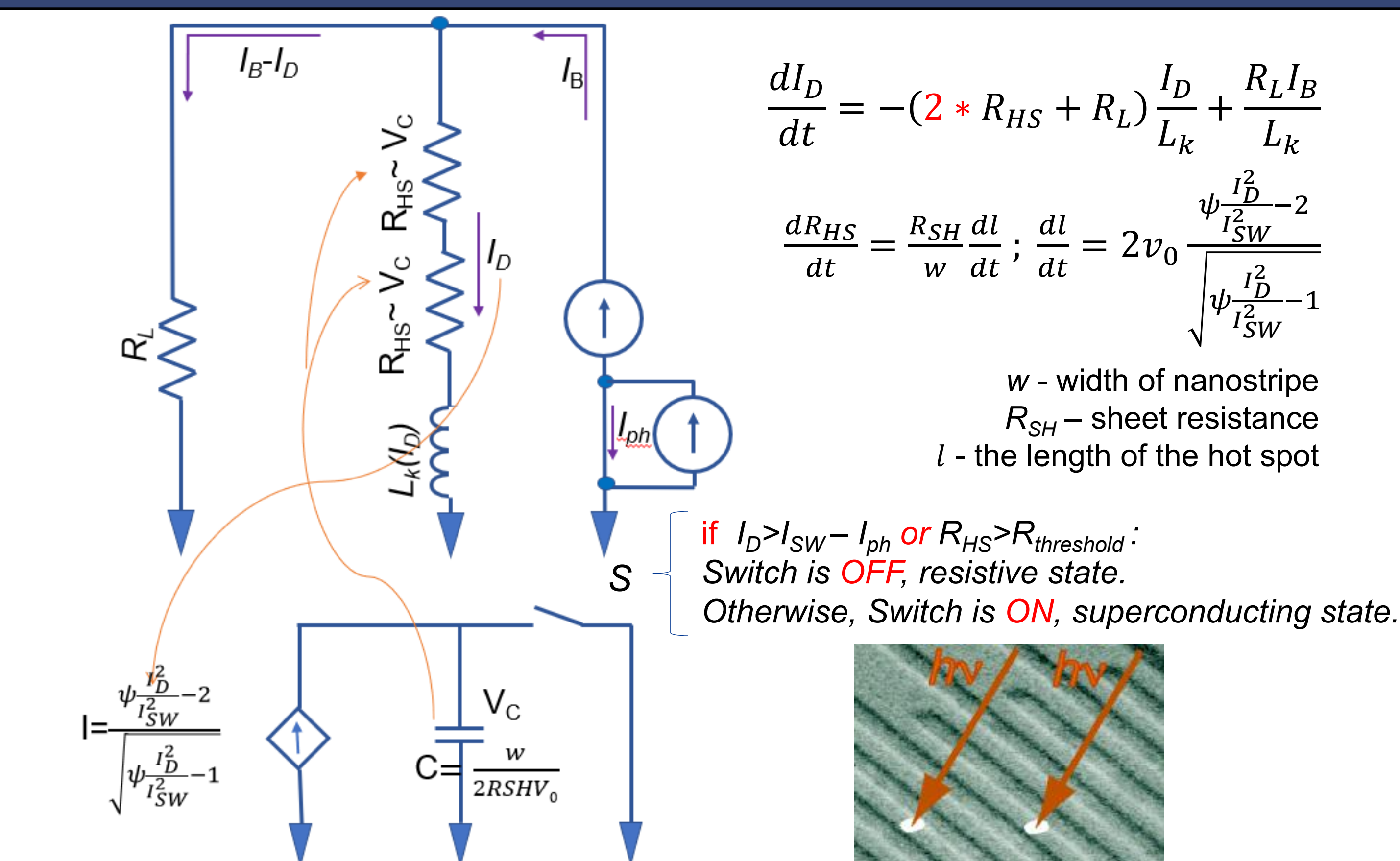


FIG. 2: Schematic diagram of the SNSPD Dynamic Model for case of a two-photon absorption event.

We use a modified SPICE dynamic model [7] for the SNSPD photoresponse simulation of up to five photon absorption events based on equivalent number of hotspot resistances connected in series.

SPICE Simulation Results

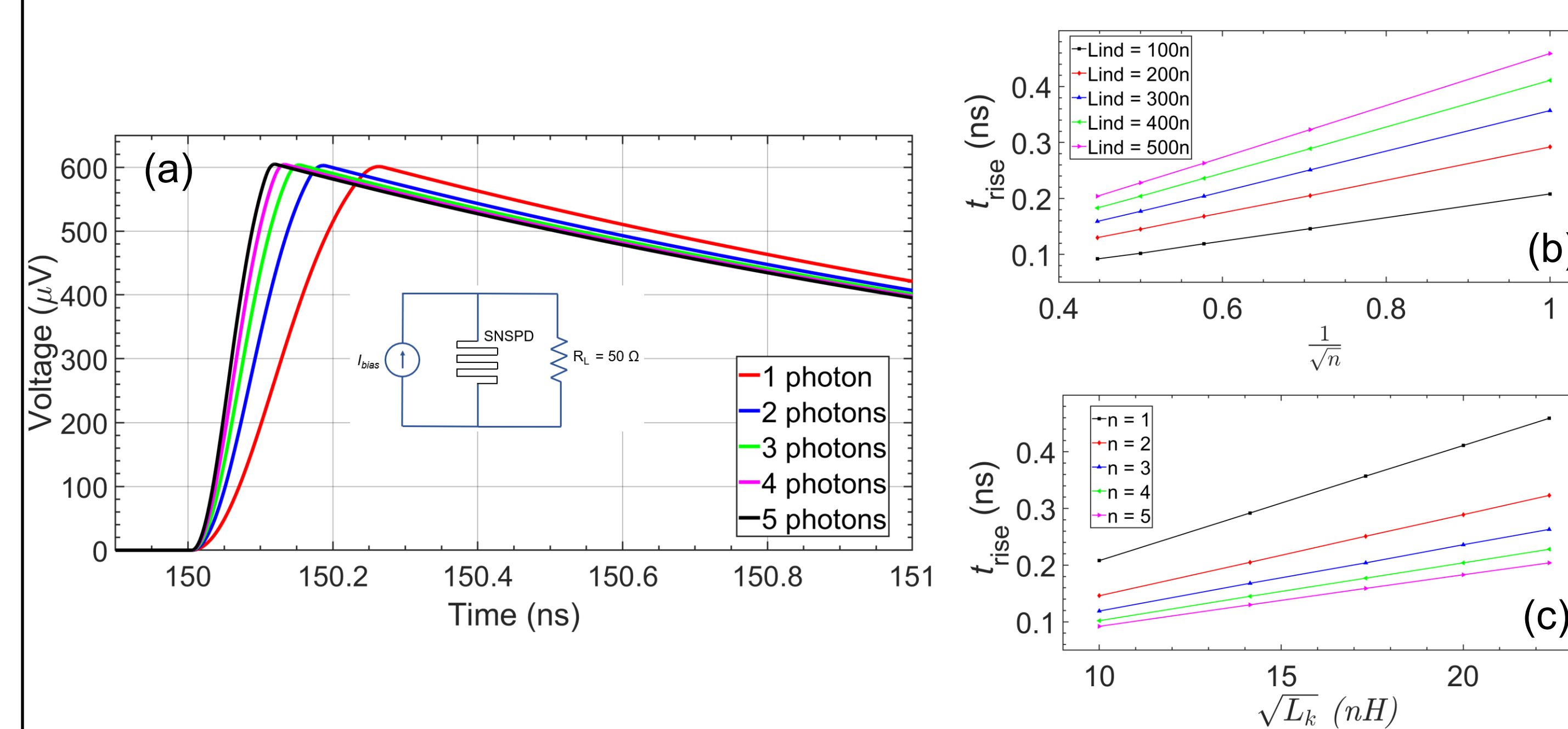


FIG. 3: (a) Plot of SNSPD pulses for 1-5 photon detection events, from voltages across R_L in the simplify photoresponse electrical model of SNSPD with $L_k = 100$ nH. (b) Photoresponse risetime in relation to the number of photons and the meander length l , for several L_k values. (c) Photoresponse risetime in relation to L_k for photon numbers from 1 to 5.

The total resistance of the nanostripe due to the presence of n hotspots is given by nR_{HS} . Under the electrothermal model of the SNSPD, risetimes of output pulses for a given number of photons and SNSPD L_k are shown to scale as $t_{rise} \propto \sqrt{\frac{L_k}{n}}$.

SNSPD kinetic inductance

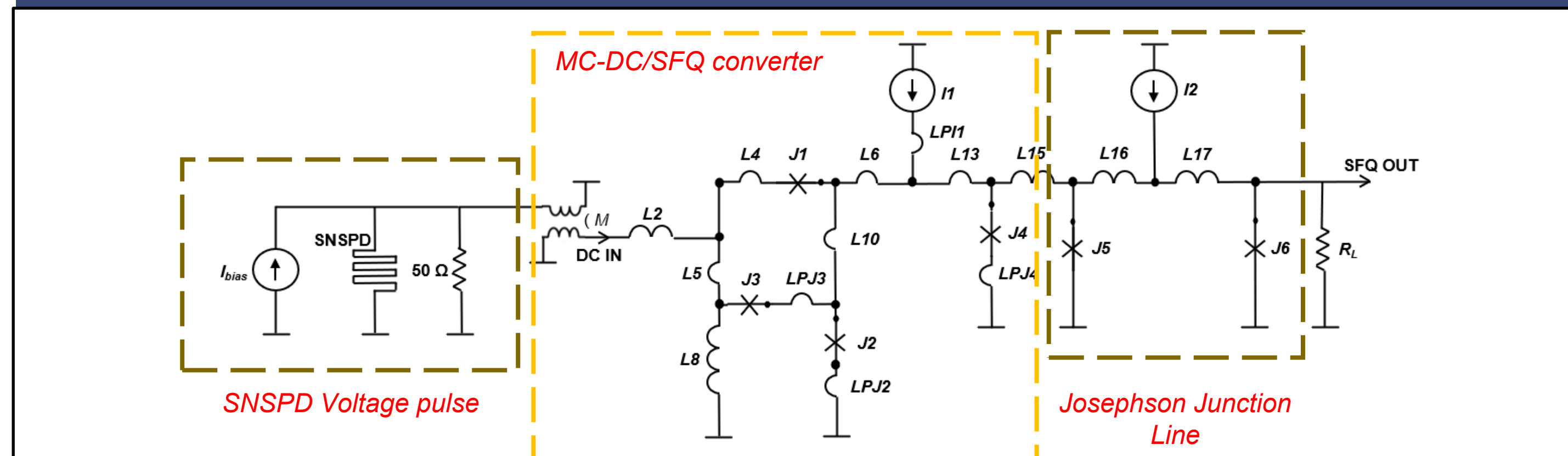


FIG. 4: Simple SFQ readout circuit to achieve timing resolution of SFQ output pulses in relation to the SNSPD kinetic inductance.

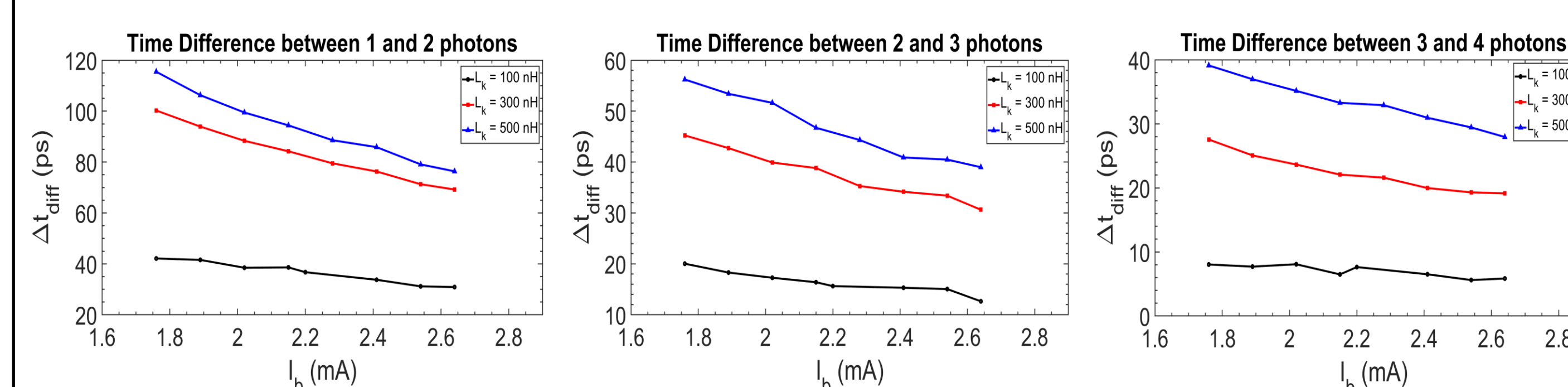


FIG. 5: Time differences between SFQ pulses corresponding to n and $n + 1$ absorbed photons in relation to the SNSPD bias current, for several L_k values.

Adjusting the L_k value of an SNSPD makes it possible to control time delay between SFQ pulses corresponding to different photon-number events. The current bias margins are within $\pm 20\%$.

SFQ Readout for Photon Number Resolution

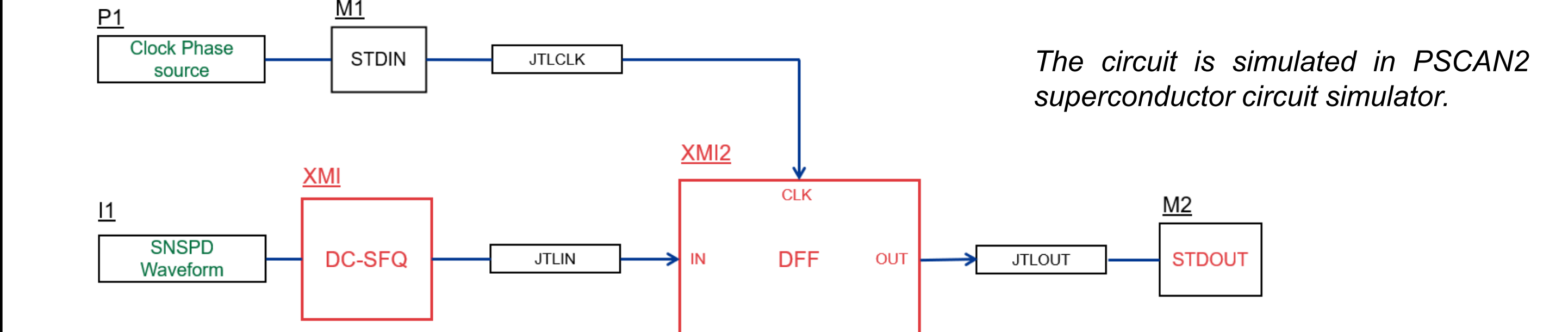


FIG. 6: SFQ-based PNR readout scheme for a large kinetic inductance SNSPD.

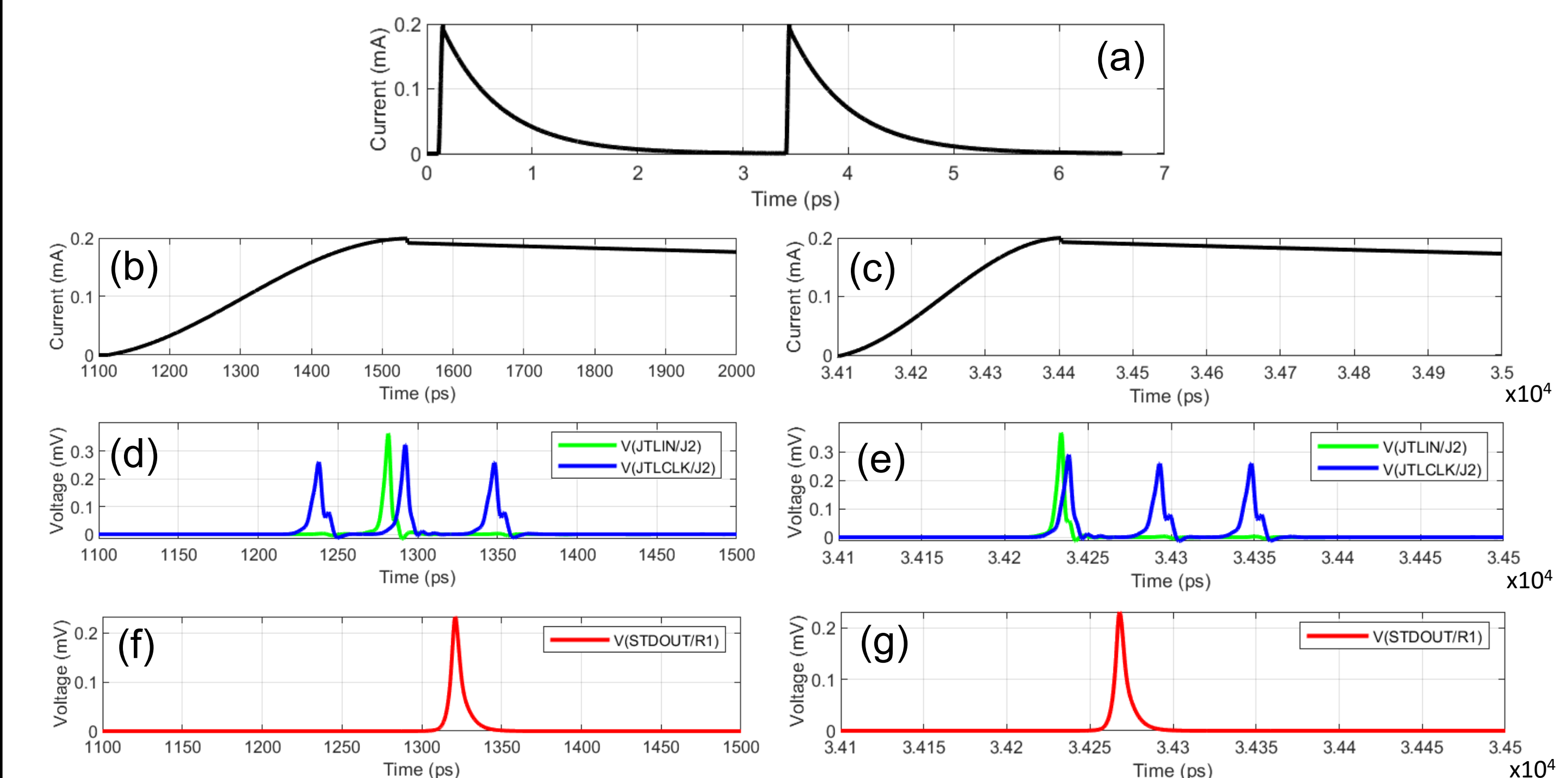


FIG. 7: (a) Sequences of SNSPD-generated pulses for 1- and 2-photon events. Zoomed rising front for (b) 1- and (c) 2-photon events. Superposition of DC-to-SFQ-generated pulses and clock pulses (d) 1- and (e) 2-photon events. DFF output pulse for (f) 1- and (g) 2-photon events.

Conclusions

Our simulations have demonstrated feasibility of an SFQ-based PNR readout architecture for a large-meander-type SNSPD. The modeled photoresponse risetime scales as $t_{rise} \propto \sqrt{L_k/n}$, while the SFQ readout provides easily to distinguish timing signatures for different photon-number events, together showing a clear route toward scalable, low-power, and cryo-compatible PNR detection nodes for high-rate quantum communication systems and networks.

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